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On the characterization of eventually norm continuous semigroups in Hilbert spaces

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It is an old problem to characterize eventually norm continuous semigroups through growth conditions for their resolvent. Besides the Hille-Yosida theorem for strongly continuous semigroups, known characterizations exist for nilpotent (see [4, § 6.10]), differentiable, eventually differentiable and analytic semigroups (see [3, Chapt. 2.4, 2.5], [2, A-II]).

In general Banach spaces a similar result for eventually norm continuous semigroups still seems to be unknown. However, recently P. You [5] showed that for semigroups in Hilbert spaces the norm continuity for $t > 0$ is equivalent to the decay to zero of its resolvent along some imaginary axis. The proof given in [5] is quite technical. It combines the complex inversion formula for the Laplace transform and a transformation of the integration path using the fact that the spectrum of the generator is "small", i.e. bounded on imaginary axes.

The aim of this paper is to give a much simpler and more straight forward proof of this result. Moreover, as an easy consequence we also obtain a characterization of semigroups which are norm continuous for $t > t_0$. This result is of particular interest since for eventually norm continuous semigroups the spectral and the growth bounds coincide. Therefore the asymptotic behaviour of solutions of the associated Cauchy problem is determined by the location of the spectrum of the generator. For a more detailed analysis of the asymptotic behavior of strongly continuous semigroups we refer to [2, A-III, IV].

Theorem 1 ([5, Thm. 1]). *Let A be the generator of a strongly continuous, exponentially stable semigroup $(T(t))_{t \geq 0}$ on a Hilbert space H. Then the following conditions are equivalent.*

(a) $(T(t))_{t \geq 0}$ *is norm continuous for* $t > 0$.

(b) $\lim_{\mathbb{R} \ni \mu \to \pm \infty} \| R(i\mu, A) \| = 0$.

Assertion (a) $\Rightarrow$ (b) is well-known, see e.g. [3, Chapt. 2, Cor. 3.7] but for the sake of completeness and for the proof Theorem 4 we include the following result. In particular, it implies (a) $\Rightarrow$ (b) if we choose $F = \mathcal{L}(H)$, $S(t) = T(t)$ and $\mu_0 = 0$.

Lemma 2. *Let F be a Banach space and $S : [0, \infty) \to F$ a function which is continuous for $t > 0$. Moreover assume that there exist $M > 0, \omega \in \mathbb{R}$ such that $\|S(t)\| \leqq M e^{\omega t}$. Then the Laplace transform $R(z) := \int_0^\infty e^{-zt} S(t)\, dt$ exists for all $z \in \mathbb{C}$ satisfying $\operatorname{Re} z > \omega$ and*

$$\lim_{\mathbb{R} \ni \mu \to \pm\infty} R(\mu_0 + i\mu) = 0$$

for every $\mu_0 > \omega$.

Proof. All assertions but the last one are obvious. We have $\|e^{-(\mu_0 + i\mu)t} S(t)\| \leqq M e^{(\omega - \mu_0)t}$ where $\omega - \mu_0 < 0$. Hence for $\varepsilon > 0$ there exists $N > 0$ such that $\|\int_N^\infty e^{-(\mu_0 + i\mu)t} S(t)\, dt\| < \varepsilon$. Now the Riemann-Lebesgue lemma implies that $\int_0^N e^{-(\mu_0 + i\mu)t} S(t)\, dt$ converges to zero for $\mathbb{R} \ni \mu \to \pm\infty$ and the lemma is proved. □

In order to show the converse implication we need the following representation formula for the semigroup $(T(t))_{t \geqq 0}$ which is valid on arbitrary Banach spaces.

Lemma 3. *Let A generate a strongly continuous and exponentially stable semigroup $(T(t))_{t \geqq 0}$ on some Banach space E. Then*

$$T(t)\, x = \frac{(k-1)!}{t^{k-1}} \frac{1}{2\pi} \lim_{N \to \infty} \int_{-N}^{N} e^{i\lambda t} R(i\lambda, A)^k x\, d\lambda$$

for all $k \in \mathbb{N}$, $t > 0$ and $x \in D(A^2)$.

Proof. For $k = 1$ this is just the complex inversion formula for the Laplace transform, see e.g. [2, p. 116]. Note that for $x \in D(A)$ we have

$$R(i\lambda, A)\, x = \frac{1}{i\lambda} (R(i\lambda, A)\, Ax + x)$$

in particular $\lim_{\lambda \to \pm\infty} R(i\lambda, A)\, x = 0$. Using this and integration by parts the formula easily follows by induction. □

We are now able to prove the remaining part of the theorem.

Proof of Theorem 1. (b) ⇒ (a). It suffices to show that the operator family $(t^2 T(t))_{t \geqq 0}$ is norm continuous for $t > 0$. To this end take $x \in D(A^2)$. Choosing $k = 3$ in Lemma 3 shows

$$\begin{aligned} \|t^2 T(t)\, x - s^2 T(s)\, x\| &= \frac{1}{\pi} \left\| \int_{-\infty}^{\infty} (e^{i\lambda t} - e^{i\lambda s})\, R(i\lambda, A)^3\, x\, d\lambda \right\| \\ &\leqq \frac{1}{\pi} \left\| \int_{|\lambda| \geqq N} (e^{i\lambda t} - e^{i\lambda s})\, R(i\lambda, A)^3\, x\, d\lambda \right\| \\ &\quad + \frac{1}{\pi} \int_{|\lambda| \leqq N} |e^{i\lambda t} - e^{i\lambda s}| \cdot \|R(i\lambda, A)^3\, x\|\, d\lambda \\ &=: I_1(N) + I_2(N). \end{aligned}$$

Let $\varepsilon > 0$. We show that $I_1(N) < \varepsilon \|x\|$ for N sufficiently large. To this end take $x^* \in H$.

Then by the Cauchy-Schwarz and Hölder inequalities we obtain

$$\left|\left\langle \int_{|\lambda|\geq N} (e^{i\lambda t} - e^{i\lambda s})\, R(i\lambda, A)^3 x\, d\lambda \,|\, x^* \right\rangle\right| \leq 2 \int_{|\lambda|\geq N} |\langle R(i\lambda, A)^2 x \,|\, R(i\lambda, A)^* x^* \rangle|\, d\lambda$$

$$\leq 2 \left(\int_{|\lambda|\geq N} \| R(i\lambda, A)^2 x \|^2 d\lambda \right)^{\frac{1}{2}} \cdot \left(\int_{|\lambda|\geq N} \| R(i\lambda, A)^* x^* \|^2 d\lambda \right)^{\frac{1}{2}}$$

$$\leq 2 \sup_{|\mu|\geq N} \| R(i\mu, A) \| \cdot \left(\int_{-\infty}^{\infty} \| R(i\lambda, A) x \|^2 d\lambda \right)^{\frac{1}{2}} \cdot \left(\int_{-\infty}^{\infty} \| R(i\lambda, A)^* x^* \|^2 d\lambda \right)^{\frac{1}{2}}$$

$$\leq 4\pi \sup_{|\mu|\geq N} \| R(i\mu, A) \| \cdot \| T(\cdot)\, x \|_2 \cdot \| T(\cdot)^* x^* \|_2 , \tag{1}$$

where in the last step we used the equality

$$\| T(\cdot)\, x \|_2 := \left(\int_0^{\infty} \| T(t)\, x \|^2 dt \right)^{\frac{1}{2}}$$

$$= \left(\int_{-\infty}^{\infty} \left\| \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\lambda i t}\, T(t)\, x\, dt \right\|^2 d\lambda \right)^{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} \| R(i\lambda, A) x \|^2 d\lambda \right)^{\frac{1}{2}},$$

which follows from the Plancherel theorem for the Hilbert space valued Fourier transform (see [1, Lemma 2]). Since $(T(t))_{t\geq 0}$ and the adjoint semigroup $(T(t)^*)_{t\geq 0}$ are exponentially stable there exists $M > 0$ such that

$$\| T(\cdot)\, x \|_2 \leq M \| x \| \quad \text{and} \quad \| T(\cdot)^* x^* \|_2 \leq M \| x^* \| .$$

Combining this with estimate (1) we obtain

$$\left\| \int_{|\lambda|\geq N} (e^{i\lambda t} - e^{i\lambda s})\, R(i\lambda, A)^3 x\, d\lambda \right\| = \sup_{\| x^* \| \leq 1} \left| \left\langle \int_{|\lambda|\geq N} (e^{i\lambda t} - e^{i\lambda s})\, R(i\lambda, A)^3 x\, d\lambda \,|\, x^* \right\rangle \right|$$

$$\leq 4\pi M^2 \cdot \sup_{|\mu|\geq N} \| R(i\mu, A) \| \cdot \| x \| ,$$

hence

$$I_1(N) \leq 4M^2 \cdot \sup_{|\mu|\geq N} \| R(i\mu, A) \| \cdot \| x \| .$$

Since $\lim_{\mathbb{R}\ni\mu\to\pm\infty} \| R(i\mu, A) \| = 0$ there exist $N > 0$ such that

$$4M^2 \cdot \sup_{|\mu|\geq N} \| R(i\mu, A) \| < \varepsilon ,$$

which yields the desired estimate $I_1(N) < \varepsilon \| x \|$ for every $x \in D(A^2)$.

In order to estimate $I_2(N)$ choose $\delta \in (0, t)$ such that

$$|e^{i\lambda t} - e^{i\lambda s}| \cdot \| R(i\lambda, A)^3 \| < \frac{\varepsilon}{2N}$$

for all $s \in (t-\delta, t+\delta)$, $\lambda \in [-N, N]$. Then $I_2(N) < \varepsilon \|x\|$ for all $x \in H$. Using these estimates for $I_1(N)$, $I_2(N)$ and the fact that $D(A^2)$ is dense in H yields

$$\| t^2 T(t) - s^2 T(s) \| < 2\varepsilon$$

for all $s \in (t-\delta, t+\delta)$. This completes the proof. □

Remark. In the proof of Theorem 1 we used Lemma 3 for $k = 3$. Applying it for $k = n+2$ some easy changes in our proof show that the condition

$$\text{(b')} \qquad \lim_{\mathbb{R} \ni \mu \to \pm\infty} \| R(i\mu, A)^n \| = 0 \text{ for some } n \in \mathbb{N}$$

also implies the norm continuity of $(T(t))_{t \geq 0}$ for $t > 0$. In particular, it follows that condition (b) of Theorem 1 and condition (b') are equivalent.

Clearly the assumption of exponential stability of the semigroup is not essential. Otherwise consider $A - \alpha$ for some $\alpha > \omega(A)$ instead of A. We can also generalize Theorem 1 to yield a characterization of eventually norm continuous semigroups on Hilbert spaces.

Theorem 4. *Let A be the generator of a strongly continuous semigroup $(T(t))_{t \geq 0}$ on a Hilbert space H. Then the following conditions are equivalent.*

(a) *$(T(t))_{t \geq 0}$ is norm continuous for $t > t_0$.*

(b) *There exist $\mu_0 > \omega(A)$ and $n \in \mathbb{N}$ such that* $\lim_{\mathbb{R} \ni \mu \to \pm\infty} \| R(\mu_0 + i\mu, A)^n T(t_0) \| = 0$.

Proof. (a) $\Rightarrow$ (b) for $n = 1$ follows from Lemma 2 if we choose $F = \mathcal{L}(H)$, $S(t) = T(t + t_0) = T(t)\, T(t_0)$ and some $\mu_0 > \omega(A)$. The converse implication follows after some obvious changes of the proof of Theorem 1. In fact, one obtains an estimate like (1) where $R(i\mu, A)$ is replaced by $R(i\mu, A)^n\, T(t_0)$. □

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